

FA7 Part 2

Diche, Jasmine & Hermosada, Joshua

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```
library(readxl)
lunch_money <- read_excel("fa7.xlsx")
names(lunch_money)
```

```
## [1] "Student"      "Lunch Money"
```

```
mean(lunch_money$Lunch Money, na.rm = TRUE)
```

```
## [1] 165
```

```
sd(lunch_money$Lunch Money, na.rm = TRUE)
```

```
## [1] 66.43241
```

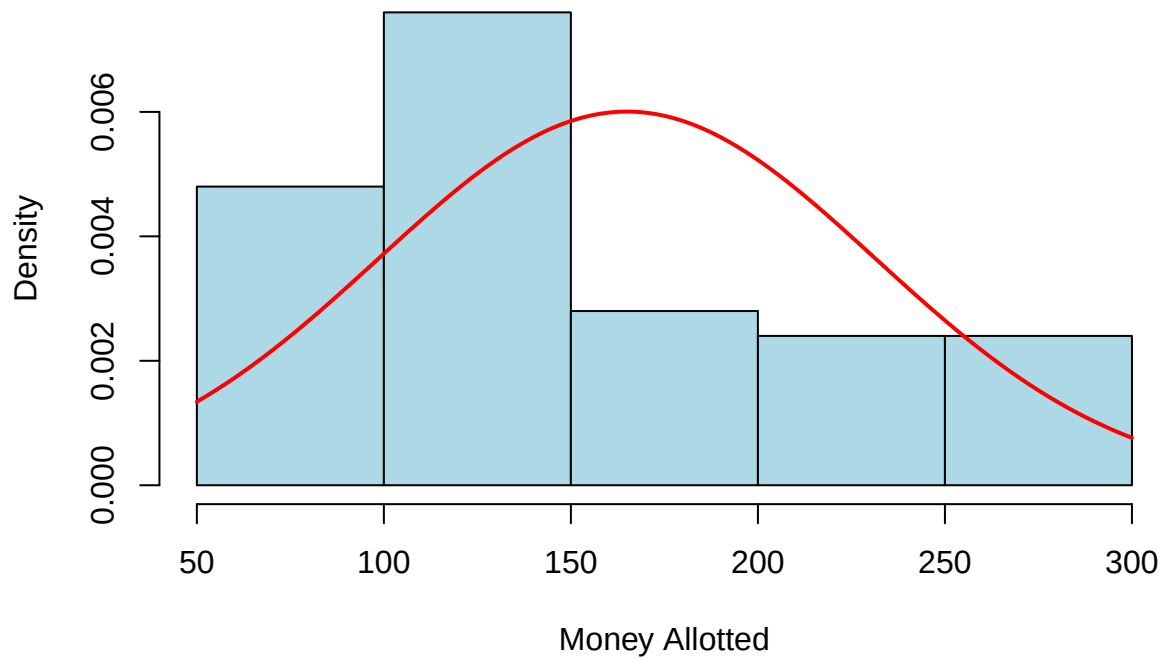
```
table(cut(lunch_money$Lunch Money, breaks = 5))
```

```
##
## (79.8,120] (120,160] (160,200] (200,240] (240,280]
##          22          9          7          0          12
```

```
# Plotting
```

```
hist(lunch_money$Lunch Money, probability = TRUE, col = "lightblue", main = "Normal Distribution of Lunch Money")
curve(dnorm(x, mean = 165, sd = 66.43), add = TRUE, col = "red", lwd = 2)
```

Normal Distribution of Lunch Money



```
money <- lunch_money$`Lunch Money`  
m <- 165  
s <- 66.43  
  
# 1 Standard Deviation  
cat("Percentage within 1 SD:", mean(money >= (m - s) & money <= (m + s)) * 100, "%\n")
```

```
## Percentage within 1 SD: 68 %
```

```
# 2 Standard Deviations  
cat("Percentage within 2 SD:", mean(money >= (m - 2*s) & money <= (m + 2*s)) * 100, "%\n")
```

```
## Percentage within 2 SD: 100 %
```

```
# 3 Standard Deviations  
cat("Percentage within 3 SD:", mean(money >= (m - 3*s) & money <= (m + 3*s)) * 100, "%\n")
```

```
## Percentage within 3 SD: 100 %
```